

UEI306: NETWORK ANALYSIS				
	L	T	P	Cr
	2	1	0	2.5
Course Objective: To make the students understand the concepts of graph theory, two port networks, filter design, attenuators, oscillator and network synthesis.				
Graph Theory: Graph, Tree and link branches, Network matrices and their relations, Choice of linearly independent network variables, Topological equations for loop current and for nodal voltage, Duality. Network Theorems: Superposition Theorem, Thevenin's theorem, Norton's theorem, and Maximum power transfer theorem as applied to A.C. circuits, Tellegen's theorem and their applications. Two Port Networks: Two port network description in terms of open circuits impedance, Short circuit admittance, Hybrid and inverse hybrid, ABCD and inverse ABCD parameters, Inter-connection of two port network, Indefinite admittance matrix and its applications. Network Functions: Concepts of complex frequency, Transform impedance, Networks function of one port and two port network, concepts of poles and zeros, property of driving point and transfer function. Passive Network Synthesis: Introduction, Positive Real Functions: Definition, Necessary and sufficient conditions for a function to be positive real, Synthesis vs. analysis, Elements of circuit synthesis, Foster and Cauer forms of LC Networks, Synthesis of RC and RL networks. Filters and Attenuators: Classification of filters, Analysis of a prototype low pass, High pass, Band pass, Band asymmetrical. Active Filters: Introduction to Active filters, first and second order low pass Butterworth filter, First and second order high pass Butterworth filter, Band pass filter.				
Course Learning Objectives (CLO)				
After the completion of the course, the student will be able to: 1. Apply various laws and theorems to solve electric networks. 2. Analyse the behaviour of two port networks. 3. Apply graph theory concept to solve electrical networks 4. Realise one-port network parameters 5. Design different filter and attenuator configurations.				
Text Books:				

1. Engineering Circuit Analysis, Hayt, W., Tata McGraw–Hill (2006).
2. Networks and Systems, Hussain, A., CBS Publications (2004).
3. Network Analysis, Valkenberg, Van, Prentice–Hall of India Private Limited (2007).
4. Op-Amps and Linear Integrated Circuits, Gayakwad, A. Prentice–Hall of India (2006).

Reference Books:

1. Circuit Theory, Chakrabarti, A., Dhanpat Rai and Co. (P) Ltd. (2006).
2. Networks and Systems, Roy Chowdhury, D., New Age International (P) Limited, Publishers (2007).

Evaluation Scheme	
Evaluation Elements	Weightage %
Mid Semester Test (MST)	25-30
End Semester Examination (ESE)	40-45
Sessional (may include Assignment, Sessional (Includes Regular Lab assessment) and Quizzes Project (Including report, presentation etc.)	30